

DS8003 - Mgt of Big Data and Tools

PROJECT REPORT ON AIRLINE INDUSTRY

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December 1, 2024

INTRODUCTION

The airline industry generates an enormous volume of data daily, encompassing flight schedules, operational routes, passenger demographics, preferences, delays, cancellations etc. Therefore, opportunities exist to explore these aspects from the subject dataset on the Airline industry to derive actionable insights using big data technologies. This project aims to leverage big data tools and techniques to analyze the comprehensive airline dataset, enabling us to address critical problems and challenges and improve future decision-making within the industry. The airline dataset chosen for this project includes a rich set of attributes such as flight details (flight numbers, departure and arrival times, and destinations), passenger data (age, gender, and preferences), and performance metrics (on-time arrivals, cancellations, and delays) etc. By applying big data technologies, such as distributed processing frameworks (e.g., Hadoop and Spark), data storage systems such as HDFS, and finally visualization tools such as Zeppelin, this project seeks to use and understand the power of big data analytics in solving problems and generate useful insights related to the airline industry. The insights gained from this analysis can help airlines enhance operational efficiency, improve the passenger experience, and make data-driven decisions to attain competitive advantages in the industry.

PROBLEM DEFINITION

Maintaining operational efficiency while maximizing customer satisfaction is an increasingly difficult balance in the highly competitive airline industry. Even minor disruptions, such as flight delays and cancellations, can be extremely costly for airlines and significantly impact customer perception. Passengers face the stress of potential stranding and financial penalties, while airlines risk fines and operational challenges. Moreover, flight delays contribute to environmental hazards by increasing fuel consumption and carbon emissions, as planes idle longer on the ground or in the air.

This project aims to identify the root causes of flight delays and cancellations using a big data approach, focusing on passenger demographics, flight routes, and operational factors. The goal is to optimize flight performance and enhance the customer experience by providing actionable insights.

The study will analyze which airports, countries, or continents are most prone to delays and cancellations and how passenger demographics - such as age, gender, and nationality-correlate with

travel destinations. This includes identifying frequently traveled airports and regions, examining whether the number of flights from an airport impacts cancellations, and exploring any seasonal trends in flight disruptions.

Our analysis identifies five critical aspects of airline operations for optimizing efficiency and customer satisfaction: flight delays, route planning, passenger preferences, international travel patterns, and airport performance. Through our integrated data pipeline, these interconnected elements work together to address key challenges and create opportunities. Flight delays are managed through predictive tools and pattern analysis, while route planning optimizes profitability through data-driven decisions. Understanding passenger preferences enables personalized services and targeted marketing, international travel patterns inform strategic partnerships and scheduling, and airport performance analysis helps streamline operations. By consolidating data across these domains, our project establishes a comprehensive system for continuous optimization in the competitive airline industry, facilitating both immediate improvements and long-term strategic planning.

DATASET DESCRIPTION

The Airline Dataset offers vital information on flight operations, passengers' demographics, and flight statuses. It is possible to look at the delay, the cancellation, and the passenger's behavior, making such analyses very useful for airlines to improve the operations with the optimal strategies. Also, the dataset can be utilized by regulatory as well as researchers for trend analysis, policy evaluation, and sustainability strategies improvement.

Key Attributes:

Passenger ID: Unique identifier for each passenger.

First Name: First name of the passenger.

Last Name: Last name of the passenger.

Gender: Gender of the passenger.

Age: Age of the passenger.

Nationality: Nationality of the passenger.

Airport Name: Name of the airport where the passenger boarded.

Airport Country Code: Country code of the airport's location.

Country Name: Name of the country where the airport is located.

Airport Continent: Continent where the airport is situated.

Continents: Continent details involved in the flight route.

Departure Date: Date when the flight departed.

Arrival Airport: Destination airport of the flight.

Pilot Name: Name of the pilot operating the flight.

Flight Status: Current status of the flight (e.g., on-time, delayed, canceled).

Additional Attributes Added

During data cleaning and transformation, the following derived attributes were introduced to enhance analysis:

- *is_delayed*: Binary indicator (1 for delayed flights, 0 otherwise).
- *is_cancelled*: Binary indicator (1 for canceled flights, 0 otherwise).
- *is_on_time*: Binary indicator (1 for on-time flights, 0 otherwise).
- Year – for partitioning
- Month - for partitioning

Dataset Size

- Rows: 98,619
- Columns: 15 original + 5 derived attributes.
- Source: Kaggle - Airline Dataset

<https://www.kaggle.com/datasets/iamsouravbanerjee/airline-dataset/data>

Data Types

- Categorical: Gender, Nationality, Flight Status, Airport Continent.
- Numerical: Age.
- Temporal: Departure Date.
- Text: Passenger and airport details (e.g., First Name, Last Name, Airport Name).

Statistics of Data

Metric	Value
Total Rows	98,619
Minimum Age	1 Year
Maximum Age	90 Years
Average Age	45.5 Years
Standard Deviation of Age	25.9 Years
Gender Distribution (count)	Male: 49,598 and Female: 49,021
Flight Status	Unique: 3 (Cancelled, On Time, Delayed)
	- Cancelled: 32,942
	- On Time: 32,846
	- Delayed: 32,831
Unique Airports	9062
Unique Pilots	98605
Departure Dates	Null values: 0 (Complete Data) Earliest: 2022-01-01 Latest: 2022-12-30
Null Arrival Airports	0 (Complete Data)
Null Flight Status	0 (Complete Data)

Limitations

- Static Data: The dataset reflects a specific snapshot in time without real-time updates.

WORK DISTRIBUTION

Name of Members	Tasks
Eyasu Deresa	Solution Architecture, Problem Definition, Dataset Description, Flight Status Analysis, International Travel Flow Analysis, and presentation deck preparation
Muhammad Maqsud Hussain	Introduction, Geographic Route Optimization Analysis, Future Works, Project document formatting and finalization
Anusri Das	Passenger Demographics Analysis, solution description and references
Beha Shirshekar	Airport Performance analysis, tools used and presentation deck preparation

SOLUTION DESCRIPTION

Tools Used

This section summarizes the tools used and justify their selection for the project. The focus here is to explain why each tool was chosen and how it aligns with the project's requirements. More technical details, including implementation steps and configurations, are presented in the Solution Architecture section.

1. HDFS (Hadoop Distributed File System):

HDFS is designed to handle massive datasets with scalability, fault tolerance, and distributed storage. Its ability to store data across multiple nodes in a cluster made it an ideal choice for this big data project.

For this project, HDFS Served as the primary storage system for both raw and processed data, ensuring redundancy and reliability for large-scale data processing.

2. Hive:

Hive provides a SQL-like interface for querying and managing large datasets stored in HDFS. It was chosen for its simplicity and compatibility with structured data processing. Also, it reduces storage

costs through data compression and partitioning. Excellent for ETL.

Hive is used in this project Used for data cleaning, applying transformations, and preparing data with features like partitioning and bucketing to enhance query performance.

3. Spark SQL:

Spark SQL was selected for its in-memory processing capabilities, which deliver significantly faster performance than traditional batch processing frameworks like MapReduce or Hive. And it's selected for performance advantages, integration with Python, Zeppelin, and better support for machine learning workflows.

Enabled efficient querying and analytical computations on the processed dataset, supporting tasks such as route optimization and delay analysis.

4. PySpark:

PySpark is chosen for its Python-based API, which allows seamless integration with Python libraries and simplifies the implementation of advanced data processing tasks. This tool provides native support for multiple data formats like csv, parquet, etc.

In this project we used parquet file format and for processing power and performance due to Lazy evaluation that optimizes the execution plan.

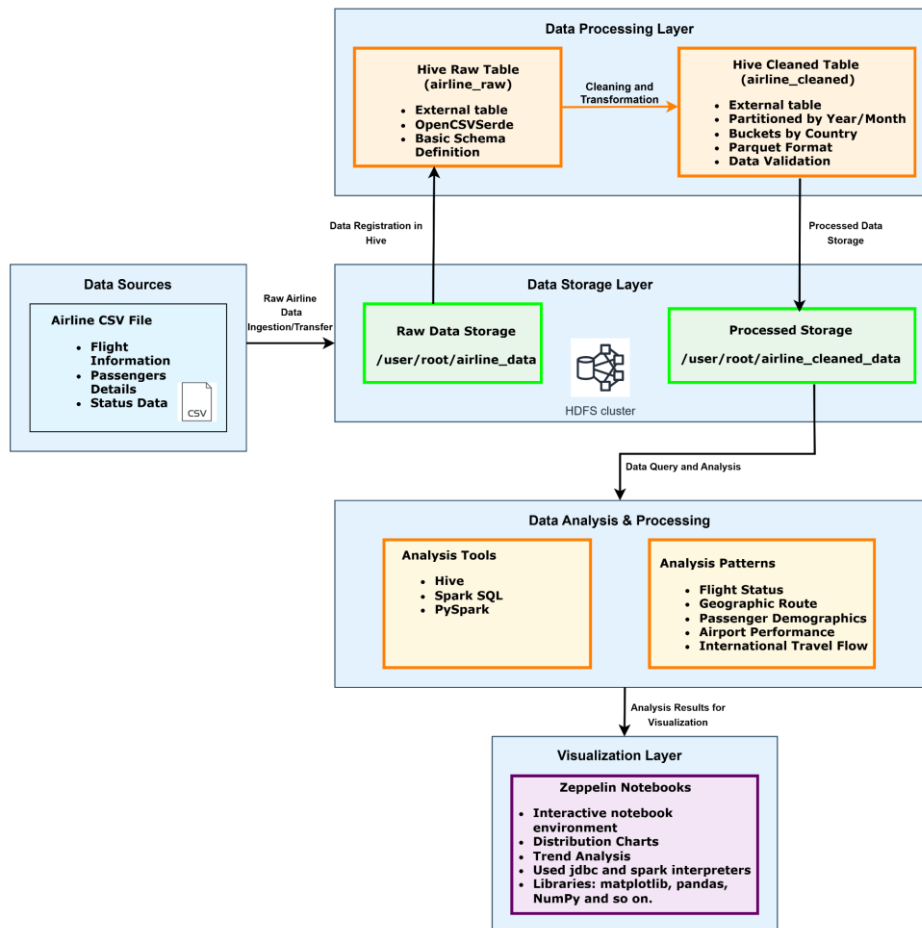
5. Zeppelin:

Zeppelin provides an interactive notebook environment that supports multiple languages, making it a versatile tool for exploratory data analysis and visualization.

In this project, we used this tool to visualize insights, enabling users to interactively explore data patterns and trends.

Solution Architecture

The Solution Architecture leverages the Hadoop ecosystem to process and analyze the Kaggle-provided airline dataset. It includes robust ingestion, storage, transformation, and optimization techniques designed to ensure efficiency and scalability for analyzing flight operations.



1. Data Ingestion

- Source: The data originates as a CSV file sourced from Kaggle.
- Process:
 - a) The CSV file is uploaded to the local file system in the HDP Virtual Machine environment using FileZilla.
 - b) From the local file system, the data is transferred to the Hadoop Distributed File System (HDFS) using the put command:
- Objective: Ensure that the dataset is accessible in the distributed storage system for subsequent processing and analysis.

2. Data Storage

- Tools: HDFS (Hadoop Distributed File System) is used for scalable and distributed storage.
- Storage Structure:
 - The raw data is stored in HDFS at /user/root/airline_data as a text file.

- Processed data is stored in HDFS at `/user/root/airline_cleaned_data` as a partitioned and bucketed Parquet file for optimized querying and retrieval.
- HDFS ensures redundancy, fault tolerance, and scalability, critical for big data applications.

3. Data Cleaning and Transformation

- Framework: Apache Hive is used for data cleaning, transformation, and preparation.
- Raw Table (`airline_raw`):
 - The `airline_raw` table stores unprocessed data in a CSV format.
 - External table configuration ensures data is not deleted when the table is dropped.
- Processed Table (`airline_cleaned`):
 - The `airline_cleaned` table introduces new fields:
 - `is_delayed`: 1 if the flight status is "Delayed"; 0 otherwise.
 - `is_cancelled`: 1 if the flight status is "Cancelled"; 0 otherwise.
 - `is_on_time`: 1 if the flight status is "On Time"; 0 otherwise.
 - Partitioning:
 - By year and month for efficient querying and analysis.
 - Bucketing:
 - By `airport_country_code` into 23 buckets to optimize join operations and query performance.
- Transformation Query: The `INSERT OVERWRITE` query extracts, cleans, and transforms the data, standardizing date formats and creating partition and bucket structures.

Data Analysis and Insights

- The cleaned and processed data in the `airline_cleaned` table is queried to generate insights on:
 1. Flight Delays: Identifying patterns and trends across airports and seasons.
 2. Route Optimization: Analyzing demand and resource utilization for route planning.
 3. Passenger Behavior: Understanding demographics to improve customer experience.
 4. International Travel Patterns: Exploring seasonal and cultural trends in cross-continental travel.
 5. Airport Performance: Evaluating operational efficiency to minimize disruptions.

5. Visualization and Reporting

Data visualizations in Zeppelin using different libraries like PySpark, Pandas, NumPy, Matplotlib etc., enhancing analytical insights.

6. Scalability and Optimization

- Partitioning and Bucketing:
 - Partitioning by year and month reduces query scope for time-based analyses.
 - Bucketing by airport_country_code optimizes joins and aggregations.

See file [hdfs_setup.sh](#) and [Basic_Schema_Definition_HIVE_Scripts.sql](#) in the source code for set up HDFS directories and for basic schema definition.

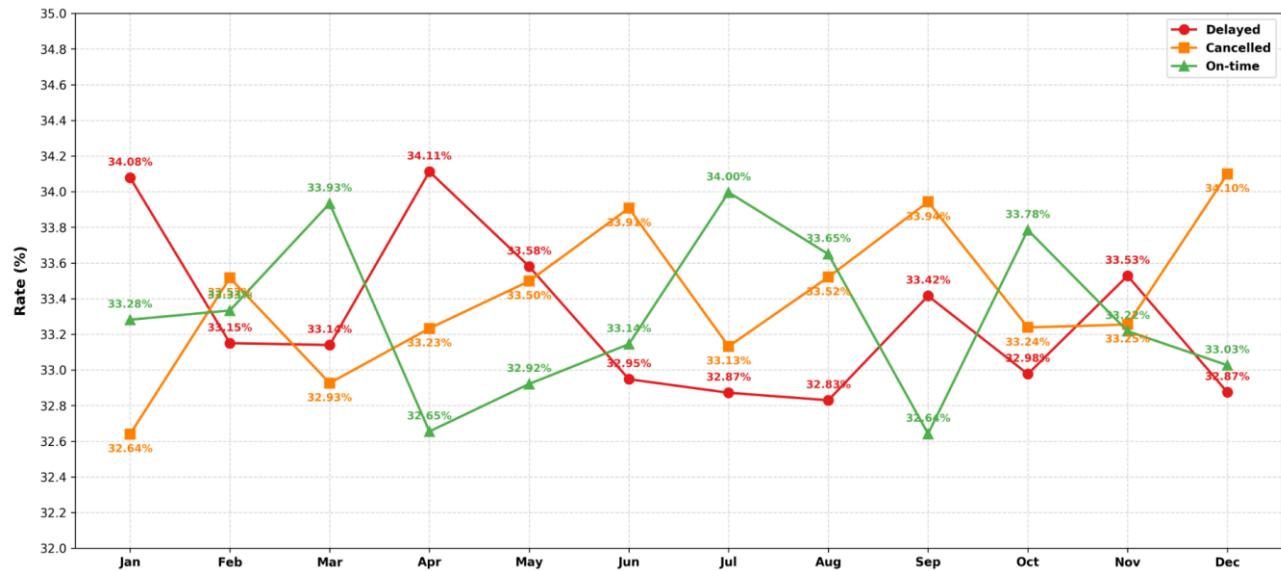
Examples of Code Snippets

```
CREATE EXTERNAL TABLE airline_cleaned (  
  passenger_id STRING,  
  first_name STRING,  
  last_name STRING,  
  gender STRING,  
  age INT,  
  nationality STRING,  
  airport_name STRING,  
  airport_country_code STRING,  
  country_name STRING,  
  airport_continent STRING,  
  continents STRING,  
  departure_date STRING,  
  arrival_airport STRING,  
  pilot_name STRING,  
  flight_status STRING,  
  is_delayed INT,  
  is_cancelled INT,  
  is_on_time INT )  
PARTITIONED BY (year STRING, month STRING)  
CLUSTERED BY (airport_country_code) INTO 23 BUCKETS  
STORED AS PARQUET  
LOCATION '/user/root/project/airline_cleaned_data';
```

The external table **airline_cleaned** stores processed airline data in Parquet format, partitioned by **year** and **month** for efficient querying. It contains passenger details, flight information, and status indicators. The data is clustered into 23 buckets by **airport_country_code** to optimize performance when filtering by country. The table is mapped to HDFS location **/user/root/project/airline_cleaned_data**. For complete implementation details, please refer to the attached source files attached with this documentation.

INSIGHTS FROM ANALYSIS

1. Flight Status Analysis



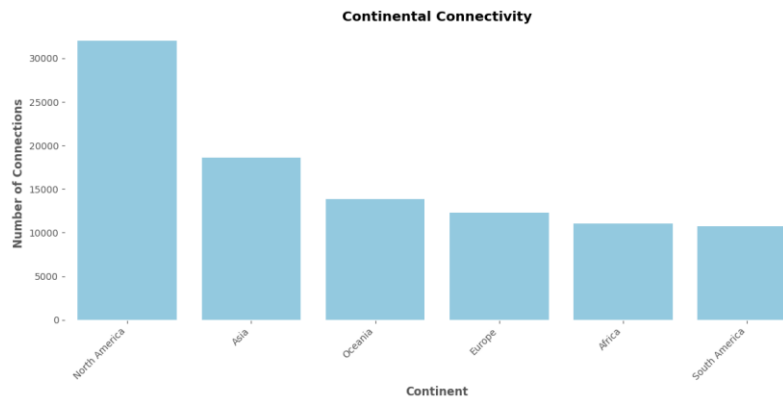
From a business perspective, the balance between these metrics significantly impacts operational costs and customer experience. The consistently high on-time performance in March (33.93%) and July (34.00%) suggests better operational efficiency during these months, potentially leading to higher customer satisfaction and reduced compensation costs. However, the concerning trend shows cancellation rates steadily increasing throughout the year, reaching a peak of 34.10% in December, which likely resulted in substantial rebooking costs and customer dissatisfaction. The delay rates show an interesting pattern, being highest in early spring (34.11% in April) and gradually improving through summer, indicating possible seasonal operational challenges. These patterns highlight the need for airlines to strengthen their winter operations, as both delays and cancellations tend to increase during these months, directly impacting the bottom line through increased operational costs and potential loss of customer loyalty. This data suggests that focusing resources on improving winter month performance could yield significant business benefits in terms of both cost reduction and customer retention.

The analysis implements two visualization scripts using PySpark and matplotlib: (1) A Flight Status Analysis script that generates a line plot showing monthly trends of delay, cancellation, and on-time rates, and (2) a Geographic Analysis script that creates a bar chart of delay rates across continents, filtered for statistical significance (>100 flights). Complete implementation details can be found in [Insight-1_FlightStatusAnalysis_Scripts.py](#)

2. Geographic Route Optimization

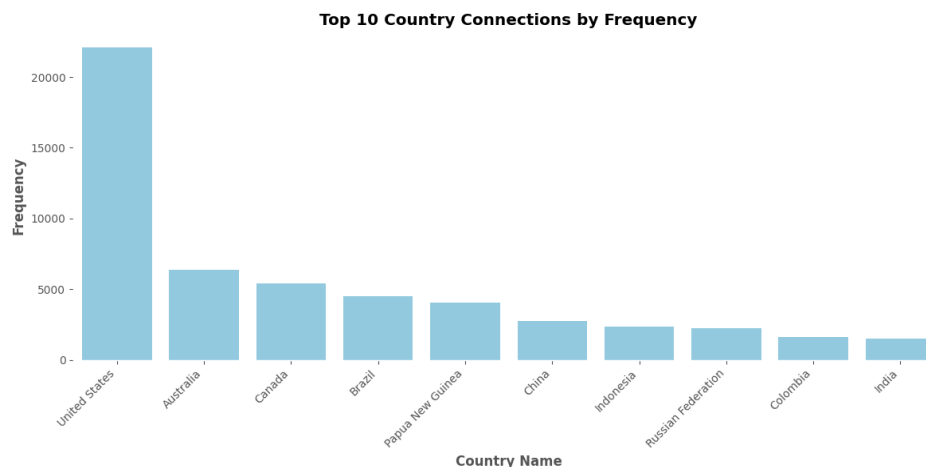
- **Continental Routes:**

As a continent, North America has the highest connections with a number of routes of 32,033. This insight transfers to the country level too where the United States is involved in the highest routes as a country and Canada is positioned third among all the nations. Asia and Oceania rank second and third as continents with route numbers of 18,637 and 13,866.



- **Country-level Connections:**

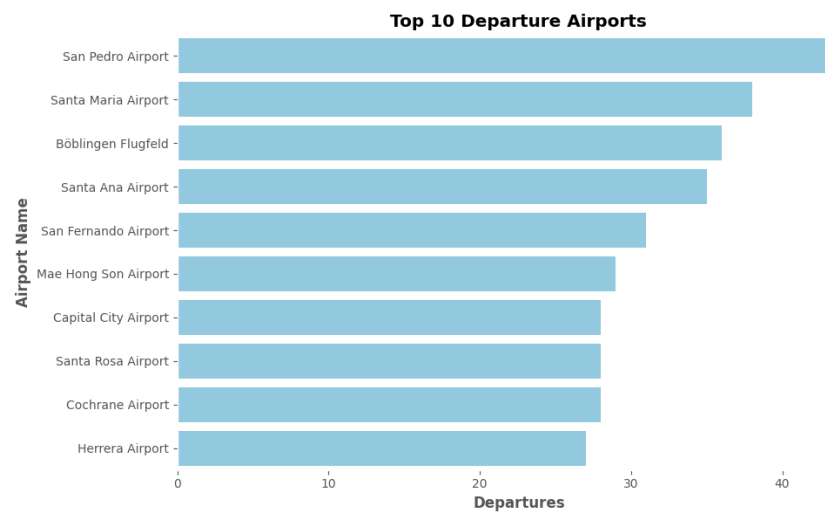
Apart from the United States with 22,104, Australia and Canada stood second with 6,370 and 5,424 routes respectively. Other major countries in the top 10 lists are Brazil, India, China etc. Therefore, the higher number of routes is commensurate with these countries' size and population. As most of these countries have a large area and/or population.



- **Airport Connectivity Patterns:**

The airport with the most departure flights *San Pedro Airport* with 43 flights and next is Santa Maria

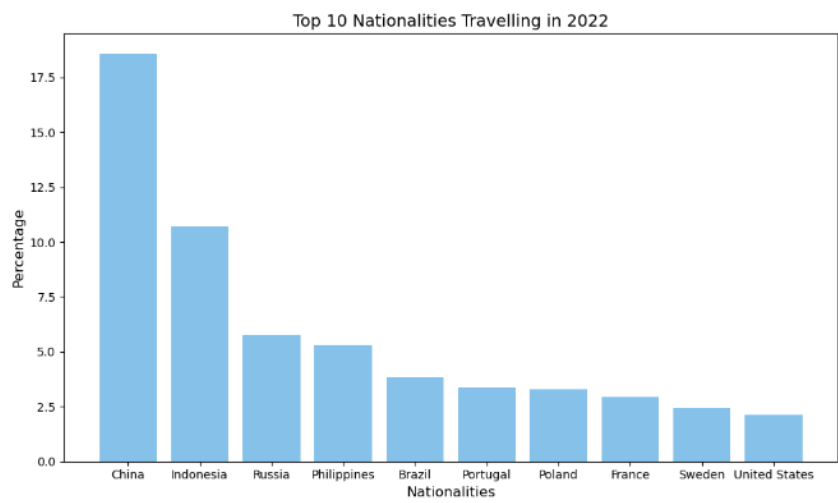
Airport with 38 flights. Both of these airports are located in North America, the continent which ranked highest in connections. However, Herrera Airport has the lowest number of departures within the list with 28 departure flights.



The analysis implements visualization scripts using PySpark and matplotlib details of which can be found in [Insight-2_Insight-2_GeographicRouteOptimization_Scripts.py](#).

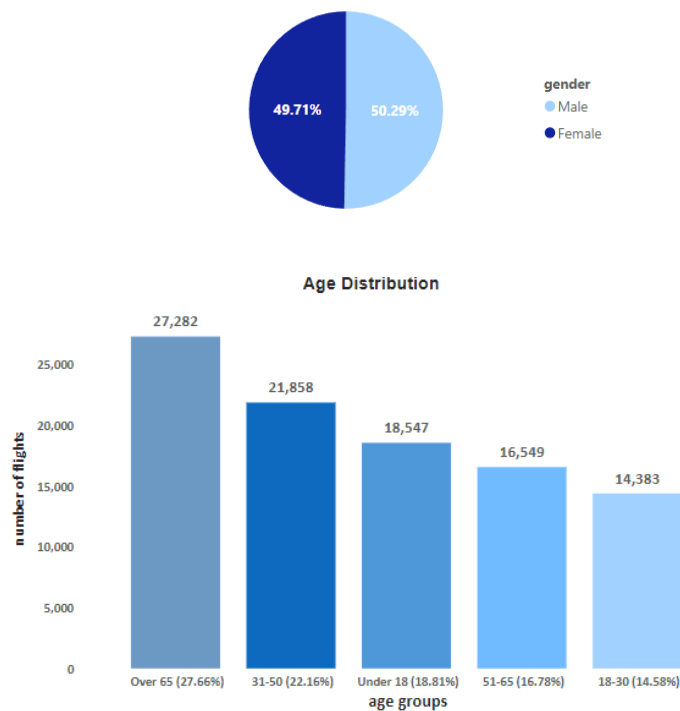
3. Passenger Demographic Analysis

This analysis provides a comprehensive understanding of the global travel landscape, revealing significant trends in traveler behavior and preference.



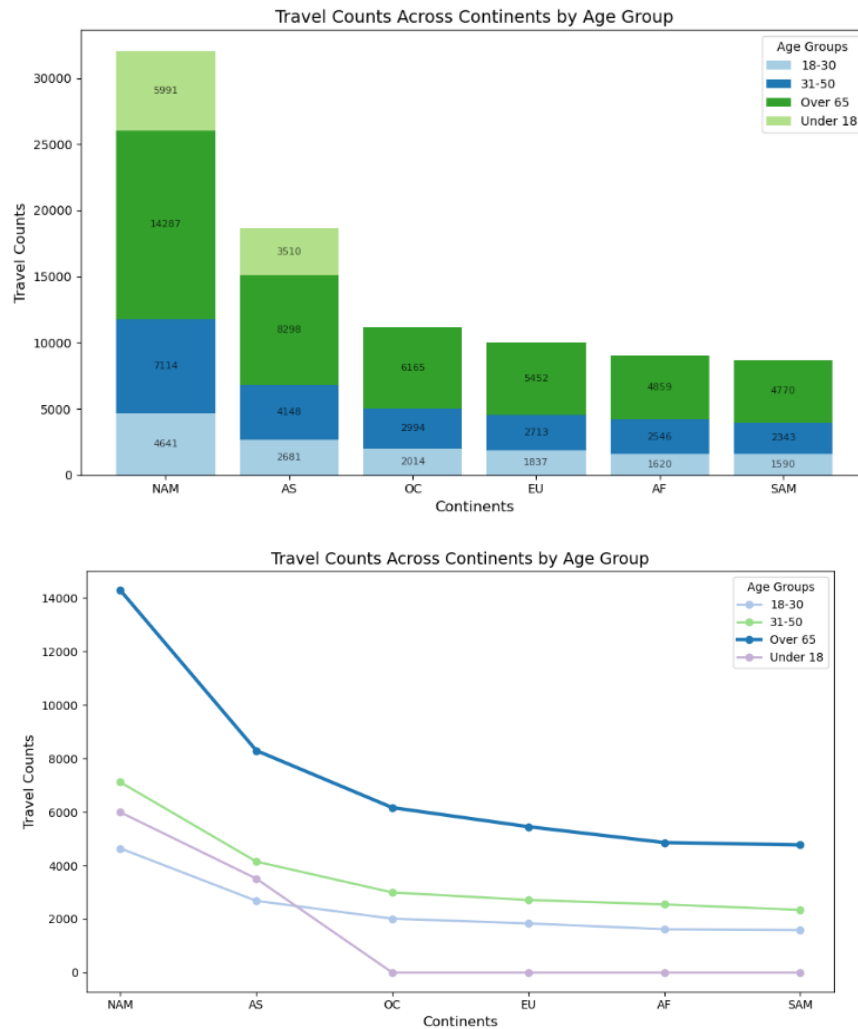
In 2022, the travel landscape was significantly influenced by national demographics, with China leading as the highest contributor of travelers at 18.67%. Following China, Indonesia accounted for 10.71%, while Russia, Philippines, and Brazil made up the third to fifth positions with 5.77%, 5.31%,

and 3.84%, respectively. This data illustrates a notable dominance of Asian and European nationalities among travelers, as the top four countries are all from Asia.



The gender distribution of passengers reflects a relatively balanced representation, as males make up 50.29% of the total passenger count, with 49,598 individuals, while females account for 49.71%, totaling 49,021 travelers. This nearly even split suggests that both genders are equally engaged in international travel.

The age distribution of passengers reveals distinct trends in travel demographics, stating that seniors (over 65) constitute the largest segment of travelers, accounting for 27.66% of the total passenger count with 27,282 individuals. This is followed by middle-aged travelers (31-50) at 22.16% and young adults (18-30) at 14.58%. Notably, children and adolescents (under 18) represent 18.81% of the total, while those aged 51-65 make up 16.78%. The data suggests potential opportunities for targeted services and marketing strategies aimed at this demographic. Understanding these age-related trends is significant for effectively catering to the diverse needs of their passengers.

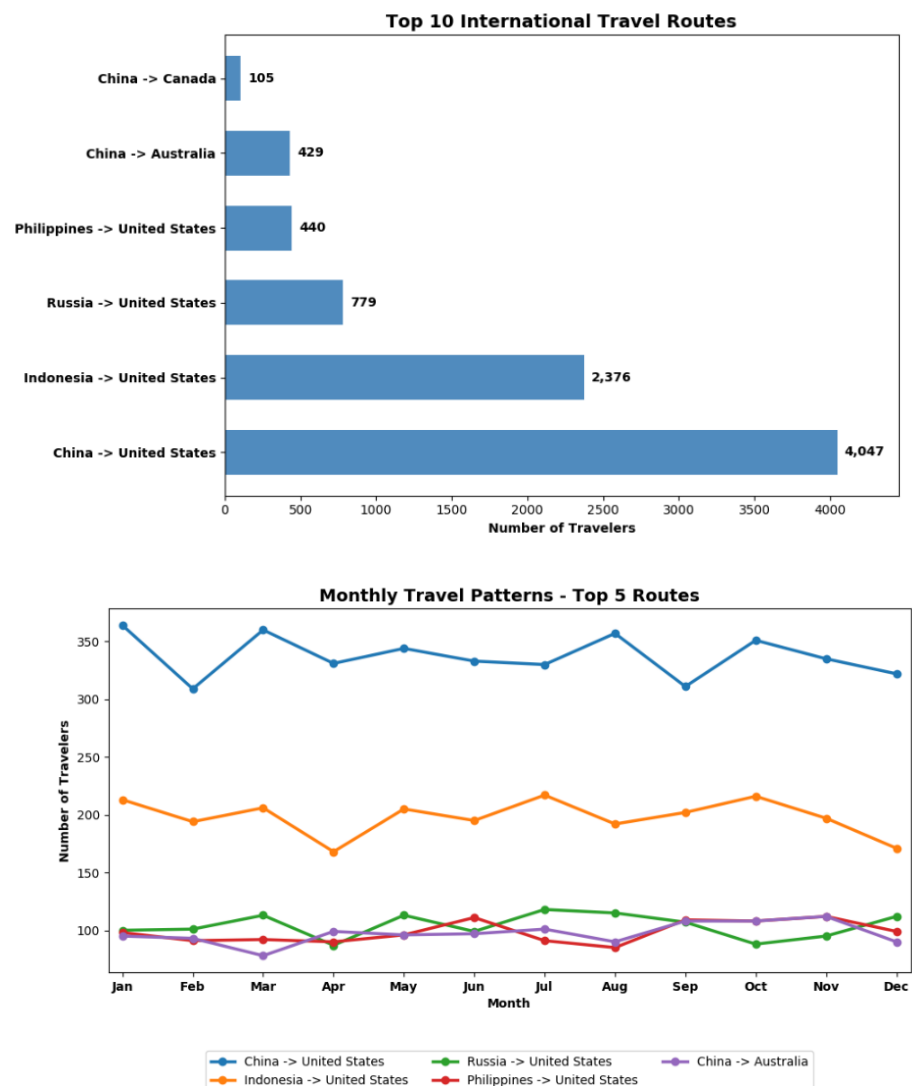


The analysis of international travel demographics shows North America as the primary destination, with seniors (65+) comprising the largest traveler segment at 27.66% (14,287 travelers), followed by adults aged 31-50 (22.16%), children under 18 (18.81%), and young adults 18-30 (14.58%). Looking at senior travel specifically, North America significantly leads other destinations, with Europe ranking second at 5,452 travelers, while Africa, Oceania, and South America show notably lower numbers. These patterns suggest opportunities for airlines to enhance senior-focused services through targeted discounts, specialized packages, improved loyalty programs, and accessibility initiatives, particularly in regions with infrastructure challenges.

The analysis implements visualization scripts using PySpark and matplotlib details of which can be found in [Insight-3_PassengerDemographicAnalysis_Scripts.py](#).

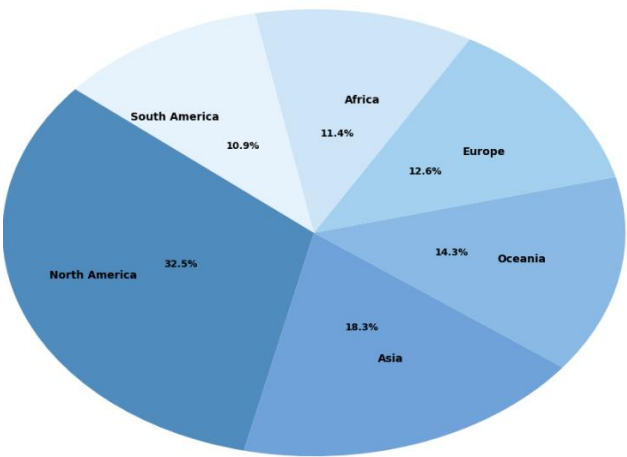
4. International Flow Travel Analysis

This analysis provides insights into the global distribution of international travel, highlighting travel flows and connectivity between continents. It helps identify key regions with high travel activity and potential areas for growth in international air traffic.



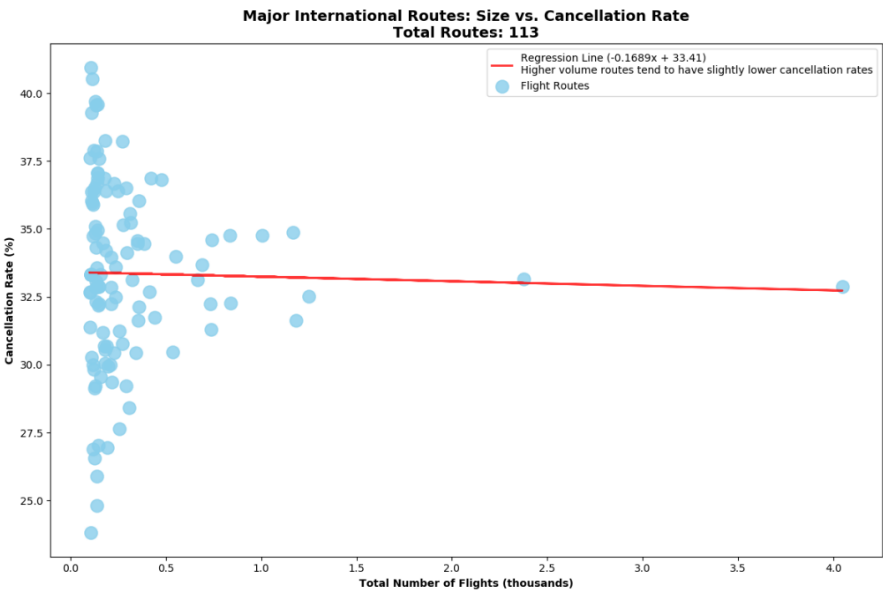
The data reveals China-United States as the dominant international travel corridor, significantly outperforming other routes in passenger volume and showing consistent year-round traffic. While this route maintains steady high volumes, other major paths like Philippines-United States and China-Australia demonstrate more variable seasonal patterns, providing valuable insights for strategic planning in areas such as resource allocation, marketing initiatives, and infrastructure development.

Distribution of International Travel by Continent:



The analysis of international travel distribution reveals North America's dominance with 32.5% of global traffic, followed by Asia at 18.3%, Oceania at 14.3%, and Europe at 12.6%. Africa and South America show lower proportions at 11.4% and 10.9% respectively, illustrating an uneven global distribution and highlighting potential growth opportunities in these regions while underscoring North America's position as a major hub for international air travel.

And related to the cancellation of flights, analysis of 113 major international routes (for those having more than 100 flights) reveals that routes with fewer flights tend to experience higher and more variable cancellation rates (up to 40%), while busier routes maintain more stable cancellation rates around 32-35%. This pattern suggests that established, high-frequency routes benefit from more robust operational systems and better resource allocation.



flight volume and cancellation rates. With a regression line of $y = -0.1689x + 33.41$, the data shows that routes with higher flight volumes tend to experience slightly lower cancellation rates - specifically, for every additional 1,000 flights on a route, the cancellation rate decreases by approximately 0.17 percentage points. While routes with very low volumes show a baseline cancellation rate of around 33.41%, the gentle downward slope suggests that more frequently traveled routes benefit from slightly better reliability, possibly due to more refined operations, experienced staff, and established infrastructure. However, the relatively small magnitude of this effect indicates that flight volume alone is not a strong determinant of cancellation rates.

This insight is valuable for the airline industry in several ways:

- Helps in resource allocation and risk assessment for new routes
- Guides strategic planning for route development and expansion
- Indicates that increasing flight frequency on routes might lead to more stable operations
- Assists in setting realistic performance targets based on route maturity and frequency
- Supports decision-making for infrastructure investment in developing routes

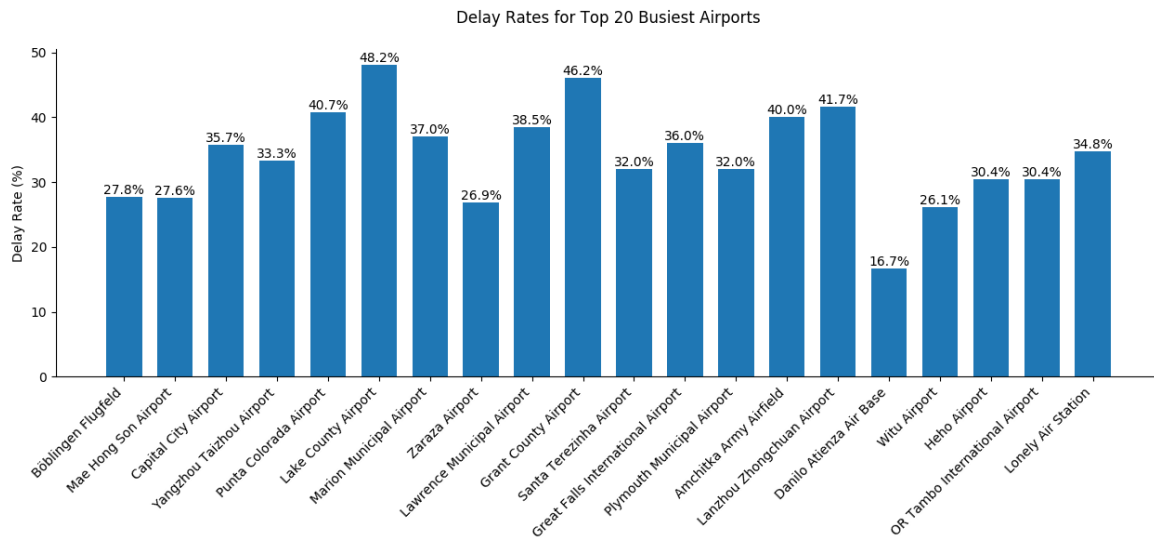
The data suggests that focusing resources on stabilizing newer or less frequent routes could help reduce their higher cancellation rates and improve overall network reliability.

Complete implementation details for this analysis can be found in the source code: [Insight-4_Insight-4_InternationalTravelAnalysis_Scripts.py](#)

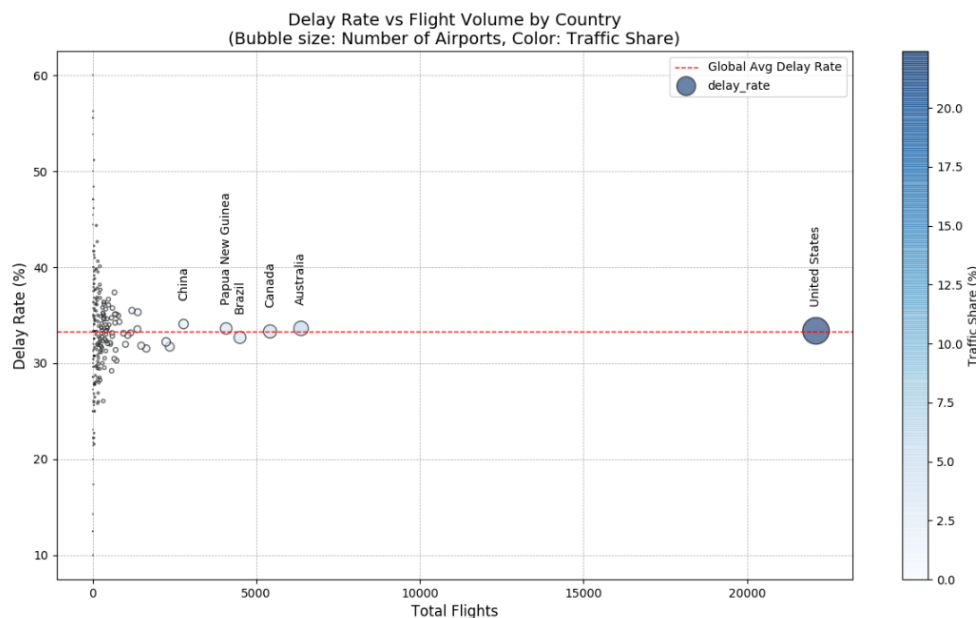
5. Airport Performance

This analysis examines the efficiency of airports worldwide by evaluating their flight volumes, delay rates, and contribution to global traffic. It identifies trends and highlights key performers to provide insights into operational efficiency and areas for improvement in airport infrastructure and management.

Airports reveal significant variations in performance across different facilities.



Analysis of airport delay rates reveals significant variations across facilities. At one extreme, Lake Country Airport leads with 48.15% delays, followed by Grant County Airport (46.15%) and Lanzhou Zhongchuan Airport (41.67%). In contrast, better-performing facilities like Witu Airport (26.09%), Del Caribe Santiago Marino International (21.74%), and Danilo Atienza Air Base (16.67%) demonstrate that sub-30% delay rates are achievable. These findings suggest that airports experiencing delays above 40% should prioritize operational improvements through infrastructure upgrades, staffing adjustments, or procedural changes, with a target benchmark of reducing delays below 30%.



The analysis reveals the United States as the dominant force in global aviation, maintaining the highest flight volume and significant traffic share while keeping delay rates near the global average, demonstrating the effectiveness of its infrastructure and flight management systems. Countries like

China, Papua New Guinea, Brazil, Canada, and Australia show moderate flight volumes with average delay rates, indicating efficient aviation operations despite their smaller scale.

Notably, some smaller countries with lower flight volumes experience higher delay rates, suggesting infrastructure and resource limitations. This data emphasizes the need for strategic infrastructure investment in underperforming regions, using high-efficiency countries like the United States as benchmarks for improving global aviation performance.

Complete implementation details for this analysis can be found in the source code: [Insight-5_Insight-5_AirportPerformanceAnalysis_Scripts.py](#)

FUTURE WORKS

The results and insights gained from this project can be taken further by integrating some additional features. The existing dataset can further be enriched with external sources such as weather data, air traffic control logs, and fuel price trends etc. This will facilitate to uncover deeper insights.

- Predictive models to forecast flight delays can be generated if weather data can be integrated along with the existing features and results from this project such as departure time and airline performance. This can be achieved by combining machine learning techniques like decision trees, random forests, or neural networks with big data tools.
- Recommendations for operational improvements can be made through optimizing flight routes based on the airport performance metrics (e.g., delay rates, cancellation rates) and by identifying underutilized connections and high-traffic routes
- Incorporating passenger satisfaction survey data to correlate service quality with performance metrics like delays, cancellations, or airport efficiency. Also, performing sentiment analysis on feedback will help to improve customer service strategies.
- Developing a real-time dashboard for monitoring flights using tools like Tableau or Power BI. This may help in incorporating live data feeds to track performance and suggest adjustments dynamically.
- Finally, it would be interesting to evaluate the profitability of routes, airlines, and airports by integrating financial data. The analysis could generate insights related to revenue and profitability which may further be expanded in recommending price adjustments or seat allocations to maximize revenue.

CONCLUSION

A comprehensive analysis of airline operations was conducted using advanced big data technologies including Hadoop, Hive, and Spark platforms. The study revealed valuable insights into flight delays, cancellations, and passenger trends while identifying key areas for operational enhancement. Future improvements could include integrating external data sources and implementing predictive analytics, ultimately leading to better decision-making and customer experience in the aviation industry through real-time monitoring and machine learning applications.

SOURCE CODES

HDFS Setup: *hdfs_setup.sh*

Basic Schema Definition (Hive): *Basic_Schema_Definition_HIVE_Scripts.sql*

Insight 1 - Flight Status Analysis: *Insight-1_FlightStatusAnalysis_Scripts.py*

Insight 2 - Geographic Route: *Insight-2_GeographicRouteOptimization_Scripts.py*

Insight 3 - Passenger Demographics: *Insight-3_PassengerDemographicsAnalysis_Scripts.py*

Insight 4 - International Travel Analysis: *Insight-4_InternationalTravelAnalysis_Scripts.py*

Insight 5 - Airport Performance Analysis: *Insight-5_AirportPerformanceAnalysis_Scripts.py*

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